LAB TASK # 04

1. Case Study:

University Database

Suppose that a database is needed to keep track of student enrollments in classes and students' final grades. After analyzing the mini world rules and the users' needs, the requirements for this database were determined to be as follows:

- The university is organized into colleges (COLLEGE), and each college has a unique name (CName), a main office (COffice) and phone (CPhone), and a particular faculty member who is dean of the college. Each college administers a number of academic departments (DEPT). Each department has a unique name (DName), a unique code number (DCode), a main office (DOffice) and phone (DPhone), and a particular faculty member who chairs the department. We keep track of the start date (CStartDate) when that faculty member began chairing the department.

- A department offers a number of courses (COURSE), each of which has a unique course name (CoName), a unique code number (CCode), a course level (Level: this can be coded as 1 for freshman level, 2 for sophomore, 3 for junior, 4 for senior, 5 for MS level, and 6 for PhD level), a course credit hour (Credits), and a course description (CDesc). The database also keeps track of instructors (INSTRUCTOR); and each instructor has a unique identifier (Id), name (IName), office (IOffice), phone (IPhone), and rank (Rank); in addition, each instructor works for one primary academic department.

- The database will keep student data (STUDENT) and stores each student's name (SName, composed of first name (FName), middle name (MName), last name (LName)), student id (Sid, unique for every student), address (Addr), phone (Phone), major code (Major), and date of birth (DoB). A student is assigned to one primary academic department. It is required to keep track of the student's grades in each section the student has completed.

- Courses are offered as sections (SECTION). Each section is related to a single or more courses and more than one instructor and has a unique section identifier (SecId). A section also has a section number (SecNo: this is coded as 1, 2, 3, . . . for multiple sections offered during the same semester/year), semester (Sem), year (Year), classroom (CRoom: this is coded as a combination of building code (Bldg) and room number (RoomNo) within the building). The database keeps track of the students in each section, and the grade is recorded when available (this is a many-to-many relationship between students and sections).

2. Case Study:

Ride Service Management System

Careem has intended to save data about rides, customers, payments, and feedback in SQL Server. This company has intelligently selected you to do the job. They want to store the information regarding the driver which includes the usual attributes like name, surname, date of birth, and driving license number. A driver drives at least one car but may drive more than one car. Each car has a model description, vehicle registration plate, owner id, and status about whether the company is still using the car. Each car is driven by exactly one driver. A car is registered in exactly one area and each area can have several registered cars. The most important information is about rides booked by the customers. Each ride is fulfilled by exactly one car. A customer can request more than one ride. The ride starts and end time, source and target location, fare, and the status of the ride (i.e., Yes/No value that tells if a ride has been canceled or accepted) should be stored efficiently. The system can keep track of drivers' working hours and total earnings per month.

3. Case Study:

Spotify

Hatfield Recruitment is a job recruitment company which has many client companies who wish them to manage the recruitment process related to job vacancies they wish to fill. Hatfield Recruitment has several hiring managers, these managers can be allocated any number of such job vacancies, from any of the client companies. Client companies are not allocated to specific hiring managers. Each job vacancy is advertised for a set period and details about the vacancy are released, such as position description, dates between which applications are to be accepted, salary range, location, and a set of prerequisites for the job, such as clean driving license, UK passport holder, etc. The hiring managers are responsible for identifying a list of suitable applicants from those applying. They will record their details such as name, dob, address and contact number. These applicants will be either existing applicants on record or new applicants. You are supposed to elaborate your use case based on brainstorming and see what attributes can each entity have if not given in case study.

Note:

- Mention attributes, entities and relationship clearly. State your assumptions.
- Do on ERD plus or Lucid chart or draw.io
- Prepare one document containing clear images of all ERDs
- Submit as i23XXXX_Lab4.pdf